

Fixed Points of a Random Permutation

Reproduction and Revised Editions

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<https://www.bilibili.com/video/BV1Fh411N71S>

Repository. The L^AT_EX sources, compiled PDFs, and edition notes are published at:

<https://github.com/Ritsuka314/personal-website>

PDF editions.

`paper-en.pdf` `paper-en-revised.pdf` `paper-zh.pdf`

Purpose. The reconstruction was prepared solely for entertainment and educational purposes. It is unofficial, is not presented as an author-approved transcript, and makes no claim of affiliation with the video creator or platform.

The three editions.

1. **Reproduced English edition.** A close typesetting of the four-page English note visible in the video. It intentionally retains the source's nonstandard wording, omissions, notation, and induction argument.
2. **Revised English edition.** A corrected, self-contained mathematical report. It defines the model precisely, proves the derangement formula by inclusion–exclusion, gives a unified probability distribution, and proves the expectation with indicator variables.
3. **Revised Chinese edition.** A polished Chinese counterpart to the revised English report. It follows the same definitions, results, and proofs rather than translating the reproduced wording literally.

Edition in this file: Reproduced English edition

This edition preserves the text and calculation visible in the source video as closely as the available frames permit.
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Lemma 1. *Consider a permutation of n elements:*

$$f : \{1, 2, \dots, n\} \longrightarrow \{1, 2, \dots, n\},$$

where f is bijective. Then the probability of getting the permutation such that

$$f(k) \neq k (\forall k \in \{1, 2, \dots, n\})$$

is

$$\sum_{i=2}^n \frac{(-1)^i}{i!}.$$

And the total number of such f is

$$n! \sum_{i=2}^n \frac{(-1)^i}{i!}.$$

The proof of this lemma is omitted here. Assume X is a random variable referring to how many elements in the permutation are corrected aligned (i.e. X = the number of k 's such that $f(k) = k$). Our problem is to calculate the probability distribution of X and its expectation.

The trivial cases are

$$P(X = n) = \frac{1}{n!},$$

and

$$P(X = n - 1) = 0.$$

In general, for $k \in \{0, 1, 2, \dots, n - 2\}$,

$$\begin{aligned} P(X = k) &= P(k \text{ elements aligned with the other } (n - k) \text{ elements ALL misaligned}) \\ &= \frac{1}{n!} \binom{n}{k} \left(\sum_{i=2}^{n-k} \frac{(-1)^i}{i!} \right) (n - k)! \\ &= \frac{1}{k!} \sum_{i=2}^{n-k} \frac{(-1)^i}{i!}. \end{aligned}$$

where $n!$ is the total number of cases considered, $\binom{n}{k}$ is the binomial coefficient C_n^k , and

$$\left(\sum_{i=2}^{n-k} \frac{(-1)^i}{i!} \right) (n - k)!$$

is the number of cases where all $(n - k)$ fixed elements are misaligned in the alignment.

Therefore, the probability distribution of X is

$$P(X = k) = \begin{cases} \frac{1}{k!} \sum_{i=2}^{n-k} \frac{(-1)^i}{i!}, & \text{if } k \in \{0, 1, 2, \dots, n - 2\}, \\ 0, & \text{if } k = n - 1, \\ \frac{1}{n!}, & \text{if } k = n. \end{cases}$$

Proposition 1. *The expectation of X is*

$$E(X) = 1.$$

Proof.

$$\begin{aligned} E(X) &= \frac{n}{n!} + (n-1) \cdot 0 + \sum_{k=0}^{n-2} \frac{k}{k!} \sum_{i=2}^{n-k} \frac{(-1)^i}{i!} \\ &= \frac{1}{(n-1)!} + \sum_{k=1}^{n-2} \frac{1}{(k-1)!} \sum_{i=2}^{n-k} \frac{(-1)^i}{i!} \\ &= \frac{1}{(n-1)!} + \sum_{k=1}^{n-2} \sum_{i=2}^{n-k} \frac{(-1)^i}{i!(k-1)!}. \end{aligned}$$

Use induction on n . Suppose we already have

$$\frac{1}{(n-1)!} + \sum_{k=1}^{n-2} \sum_{i=2}^{n-k} \frac{(-1)^i}{i!(k-1)!} = 1.$$

It follows that, for the $(n+1)$ case,

$$\begin{aligned}
& \frac{1}{n!} + \sum_{k=1}^{n-1} \sum_{i=2}^{n+1-k} \frac{(-1)^i}{i!(k-1)!} \\
&= \frac{1}{n!} + \frac{1}{2 \times (n-2)!} + \sum_{k=1}^{n-2} \sum_{i=2}^{n+1-k} \frac{(-1)^i}{i!(k-1)!} \\
&= \frac{1}{n!} + \frac{1}{2 \times (n-2)!} + \sum_{k=1}^{n-2} \sum_{i=2}^{n-k} \frac{(-1)^i}{i!(k-1)!} + \sum_{k=1}^{n-2} \frac{(-1)^{n+1-k}}{(n+1-k)!(k-1)!} \\
&= \frac{1}{n!} + \frac{1}{2 \times (n-2)!} + 1 - \frac{1}{(n-1)!} + \sum_{k=1}^{n-2} \frac{(-1)^{n+1-k}}{(n+1-k)!(k-1)!} \\
&= \frac{n^2 - 3n + 2}{2 \times n!} + 1 + \sum_{k=1}^{n-2} \frac{(-1)^{n+1-k}}{(n+1-k)!(k-1)!} \\
&= \frac{n^2 - 3n + 2}{2 \times n!} + 1 + \sum_{t=0}^{n-3} \frac{(-1)^{n-t}}{(n-t)!t!} \\
&= \frac{n^2 - 3n + 2}{2 \times n!} + 1 + \frac{1}{n!} \sum_{t=0}^{n-3} (-1)^{n-t} \binom{n}{t} \\
&= \frac{n^2 - 3n + 2}{2 \times n!} + 1 - \frac{1}{n!} (-1)^2 \binom{n}{n-2} - \frac{1}{n!} (-1)^1 \binom{n}{n-1} \\
&\quad - \frac{1}{n!} (-1)^0 \binom{n}{n} + \frac{1}{n!} \sum_{t=0}^n (-1)^{n-t} \binom{n}{t} \\
&= \frac{n^2 - 3n + 2}{2 \times n!} + 1 - \frac{1}{n!} \binom{n}{2} + \frac{1}{n!} \binom{n}{1} - \frac{1}{n!} + \frac{1}{n!} (-1+1)^n \\
&= \frac{n^2 - 3n + 2 - n(n-1) + 2n - 2}{2 \times n!} + 1 \\
&= 1.
\end{aligned}$$

By induction, the result

$$E(X) = 1$$

holds for all $n \in \mathbb{N}$.